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Minimax prediction for functional linear regression with functional responses in reproducing kernel Hilbert spaces

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ABSTRACT

In this article, we consider convergence rates in functional linear regression with functional responses, where the linear coefficient lies in a reproducing kernel Hilbert space (RKHS). Without assuming that the reproducing kernel and the covariate covariance kernel are aligned, convergence rates in prediction risk are established. The corresponding lower bound in rates is derived by reducing to the scalar response case. Simulation studies and two benchmark datasets are used to illustrate that the proposed approach can significantly outperform the functional PCA approach in prediction.

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1. Introduction

The literature contains an impressive range of functional analysis tools for various problems including exploratory functional principal component analysis, canonical correlation analysis, classification and regression. Two major approaches exist. The more traditional approach, masterfully documented in the monograph [16], typically starts by representing functional data by an expansion with respect to a certain basis, and subsequent inferences are carried out on the coefficients. The most commonly utilized basis includes B-spline basis for nonperiodic data and Fourier basis for periodic data. Another line of work by the French school [10], taking a nonparametric point of view, extends the traditional nonparametric techniques, most notably the kernel estimate, to the functional case. Some recent advances in the area of functional regression include Cardot et al. [6], Cai and Hall [4], Preda [15], Lian [13], Ait-Saidi et al. [2], Yao et al. [18], Crambes et al. [8], Ferraty et al. [11], Lian [14].

In this paper we study the functional linear regression problem of the form

$$Y(t) = \mu(t) + \int_0^1 \beta(t,s)X(s)\,ds + \epsilon(t), \tag{1}$$

where $Y, X, \epsilon \in L_2[0,1]$ and $E[\epsilon|X] = 0$, the same problem that appeared in [16,18,3,1,9]. In terms of methodology, the plan of attack we will give for (1) is most closely related to that of Crambes and Mas [9]. In this introduction, we will explain the methodology used in that paper and then the different assumptions we will make on $\beta(t,s)$.

Without loss of much generality, throughout the paper we assume $E(X) = 0$ and the intercept $\mu(t) = 0$, since the intercept can be easily estimated. The covariance operator of X is the linear operator $\Gamma = E(X \otimes X)$ where for $x, y \in L_2[0,1]$, $x \otimes y : L_2[0,1] \to L_2[0,1]$ is defined by $(x \otimes y)(g) = \langle y, g\rangle x$ for any $g \in L_2[0,1]$. Γ can also be represented by the bivariate

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function $\Gamma(s,t) = E[X(s)X(t)]$. Using the same letter Γ to denote both the operator and the bivariate function will not cause confusion in our context. We assume throughout the paper that $E\|X\|^4 < \infty$ which implies Γ is a compact operator. Then by the Karhunen–Loève Theorem there exists a spectral expansion for Γ,

$$\Gamma = \sum_{j=1}^{\infty} \lambda_j \varphi_j \otimes \varphi_j,$$

where $\lambda_j \geq 0$ are the eigenvalues with $\lambda_j \to 0$ and $\{\varphi_j\}$ are the orthonormalized eigenfunctions. Correspondingly, we have the representation $X = \sum_{j\geq 1} \gamma_j \varphi_j$ with $\gamma_j = \int X\varphi_j$. The random coefficient γ_j satisfies $E\gamma_j\gamma_k = \lambda_j I\{j = k\}$ where $I\{.\}$ is the indicator function.

By expanding β using the set of eigenfunctions, we write $\beta(t,s) = \sum_{j\geq 1} b_j(t)\varphi_j(s)$ and (1) can be equivalently written as

$$Y(t) = \sum_{j\geq 1} b_j(t)\gamma_j + \epsilon(t).$$

Multiplying both sides above by γ_j and taking expectations, we obtain $b_j(t) = E[Y(t)\gamma_j]/\lambda_j$. Given i.i.d. data (X_i, Y_i), $i = 1, \ldots, n$, $\{\lambda_j, \varphi_j\}$ can be easily estimated by $\hat{\lambda}_j$ and $\hat{\varphi}_j$ obtained from the spectral decomposition of the empirical covariance operator and $E[Y(t)\gamma_j]$ can be approximated by the corresponding sample average. Thus the estimator proposed in Crambes and Mas [9] is

$$\hat{\beta}(t,s) = \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{k} \frac{\int X_i\hat{\varphi}_j}{\hat{\lambda}_j} Y_i(t)\hat{\varphi}_j(s).$$

Note that the infinite sum over j has been truncated as some point k for regularization. One intriguing point is that there is no regularization on $Y_i(t)$ necessary, in contrast with Yao et al. [18] where Y is observed sparsely with additional noise. This can also be seen from that $b_j(t)$ is not a priori constrained in any way. The reason is that only regularization of the covariance operator, which does not depend on Y, is necessary to avoid overfitting.

Minimax convergence rates of $E\|\int \hat{\beta}(t,s)X(s)ds - \int \beta(t,s)X(s)ds\|^2$ were shown in Crambes and Mas [9]. Although no assumption is needed on b_j except that the regression operator is Hilbert–Schmidt, slowly decaying $\|b_j\|$ as j increases will lead to slow convergence rate of the regression function. Given that $\|b_j\|$'s are the coefficients of $\beta(t,s)$ in terms of the basis φ_j, which is a characteristic of the predictor, there is no a priori reason why this basis should provide a good representation of β in the sense that $\|b_j\|$ will decay fast. A more general assumption for β may be on its smoothness, which makes a reproducing kernel Hilbert space (RKHS) approach an interesting alternative. Such arguments have led to the developments in [19,5] for the scalar response models. While Crambes and Mas [9] is based on Cardot et al. [7] for scalar response models, ours is based on Cai and Yuan [5].

The rest of the article is organized as follows. In Section 2, we propose an estimator for β with an RKHS approach where the reproducing kernel and the covariance kernel are not necessarily aligned. Following Cai and Yuan [5], we say that the two kernels are perfectly aligned if they have the same sequence of eigenvalues when ordered according to the eigenvalues. We establish the minimax rate of convergence in prediction risk by deriving both the upper bound and the lower bound. In Section 3, we present some simulation studies to show that the RKHS approach could significantly outperform the functional PCA approach when the kernels are mis-aligned. This advantage is further illustrated on two benchmark datasets which show better prediction performance using our approach. We conclude in Section 4 with some discussions. The technical proofs are relegated to Appendix.

Finally, we list some notations and properties regarding different norms to be used. For any operator $\mathcal{F}$, we use $\mathcal{F}^{\mathrm{T}}$ to denote its adjoint operator. If $\mathcal{F}$ is self-adjoint and nonnegative definite, $\mathcal{F}^{1/2}$ is its square-root satisfying $\mathcal{F}^{1/2}\mathcal{F}^{1/2} = \mathcal{F}$. For $f \in L_2$, $\|f\|$ denotes its L_2 norm. For any operator $\mathcal{F}$, $\|\mathcal{F}\|_{op}$ is the operator norm $\|\mathcal{F}\|_{op} := \sup_{\|f\|\leq 1}\|\mathcal{F}f\|$. The trace norm of an operator $\mathcal{F}$ is $\mathrm{Trace}(\mathcal{F}) = \sum_k \langle (\mathcal{F}^{\mathrm{T}}\mathcal{F})^{1/2} e_k, e_k\rangle$ for any orthonormal basis $\{e_k\}$ of L_2. $\mathcal{F}$ is a trace class operator if its trace norm is finite. The Hilbert–Schmidt norm of an operator is $\|\mathcal{F}\|_{HS} = (\sum_{j,k}\langle \mathcal{F}e_j, e_k\rangle^2)^{1/2} = (\sum_j \|\mathcal{F}e_j\|^2)^{1/2}$. An operator is a Hilbert–Schmidt operator if its Hilbert–Schmidt norm is finite. From the definition it is easy to see that $\mathrm{Trace}(\mathcal{F}^{\mathrm{T}}\mathcal{F}) = \mathrm{Trace}(\mathcal{F}\mathcal{F}^{\mathrm{T}}) = \|\mathcal{F}\|_{HS}^2$. Furthermore, if $\mathcal{F}$ is a Hilbert–Schmidt operator and $\mathcal{G}$ is a bounded operator, then $\mathcal{F}\mathcal{G}$ is also a Hilbert–Schmidt operator with $\|\mathcal{F}\mathcal{G}\|_{HS} \leq \|\mathcal{F}\|_{HS}\|\mathcal{G}\|_{op}$.

2. Methodology and convergence rates

Following Wahba [17], a RKHS H is a Hilbert space of real-valued functions defined on, say, the interval [0, 1], in which the point evaluation operator $L_t : H \to R, L_t(f) = f(t)$ is continuous. By Riesz representation theorem, this definition implies the existence of a bivariate function $K(s,t)$ such that

$K(s,\cdot) \in H$, for all $s \in [0,1]$

and (reproducing property)

for every $f \in H$ and $t \in [0,1]$, $\langle K(t,\cdot), f\rangle_H = f(t)$.

The definition of a RKHS can actually start from a positive definite bivariate function $K(s,t)$ and RKHS is constructed as the completion of the linear span of $\{K(s,\cdot), s \in [0,1]\}$ with inner product defined by $\langle K(s,\cdot), K(t,\cdot)\rangle_H = K(s,t)$. To make the dependence on K explicit, the RKHS is denoted by H_K with the RKHS norm $\|\cdot\|_{H_K}$. With abuse of notation, K also denotes the linear operator $f \in L_2 \to Kf = \int K(\cdot,s)f(s)ds$. For later use, we note that H_K is identical to the range of $K^{1/2}$.

We assume that for any $t \in [0,1]$, $\beta(t,\cdot) \in H_K$. This is a smoothness assumption for $\beta(t,s)$ in the s-variable. As noted in the Introduction, smoothness assumption on the t-variable is not necessary. We estimate β via

$$\hat{\beta} = \arg\min_{\beta(t,.)\in H_K} \frac{1}{n}\sum_{i=1}^{n}\left\|Y_i - \int_0^1 \beta(\cdot,s)X_i(s)\,ds\right\|^2 + \lambda\int_0^1 \|\beta(t,\cdot)\|_{H_K}^2 dt. \tag{2}$$

We implicitly assume that the expression $\int_0^1 \|\beta(t,\cdot)\|_{H_K}^2 dt$ is valid, that is $\|\beta(t,\cdot)\|_{H_K}$ as a function of t is square integrable. This assumption on β is also more succinctly denoted by $\beta \in L^2 \times H_K$.

The following representer theorem is useful in computing the solution, whose proof is omitted since it is standard.

Proposition 1. *The solution of (2) can be expressed as*

$$\hat{\beta}(t,s) = \sum_{i=1}^{n} c_i(t)\int_0^1 K(s,u)X_i(u)\,du. \tag{3}$$

Based on the previous proposition, by plugging the representation (3) into (2), it can be easily shown that $(c_1(t),\ldots,c_n(t))^{\mathrm{T}} = (\Sigma + n\lambda)^{-1}Y(t)$ where Σ is an $n\times n$ matrix whose entries are given by $\Sigma_{ij} = \int\int X_i(s)K(s,t)X_j(t)dsdt$.

Remark 1. Throughout this section, we assume the reproducing kernel K is positive definite and the RKHS norm for H_K is used in the penalty. More generally, for practical use, we can assume $H_K = H_1 \oplus H_2$, where H_1, typically finite dimensional, is a RKHS with reproducing kernel K_1 and H_2 is a RKHS with reproducing kernel K_2, $K = K_1 + K_2$. We can then impose the penalty $\int \|P_2\beta(t,\cdot)\|_{H_K}^2 dt = \int \|P_2\beta(t,\cdot)\|_{H_2}^2 dt$, where P_2 is the projection onto H_2. Our theory and computation can be easily adapted to this more general case, but we use (2) for ease for presentation throughout the paper. In real data analysis, $H_K = \mathcal{W}_2^{per}$ is the second-order Sobolev space of periodic functions on $[0,1]$ and we use decomposition $H_K = H_1 \oplus H_2$ where H_1 contains the constant functions.

Since $\beta(t,\cdot) \in H_K$, there exists $f(t,s)$ such that $\beta(t,\cdot) = K^{1/2}f(t,\cdot)$ and $\|\beta(t,\cdot)\|_{H_K} = \|f(t,\cdot)\|$. Thus (2) can also be written as

$$\hat{f} = \arg\min_{f\in L_2[0,1]^2} \frac{1}{n}\sum_{i=1}^{n}\left\|Y_i - \int_0^1 f(\cdot,s)(K^{1/2}X_i)(s)\,ds\right\|^2 + \lambda\int_0^1\int_0^1 f^2(t,s)dsdt. \tag{4}$$

Due to the appearance of $K^{1/2}X_i$ in the expression above, this suggests that the spectral decomposition of $T := K^{1/2}\Gamma K^{1/2}$ plays an important role. Suppose the spectral decomposition of T is

$$T = \sum_{j\geq 1} s_j e_j \otimes e_j,$$

with $s_1 > s_2 > \cdots > 0$.

The following technical assumptions are imposed.

(A1) $s_j \asymp j^{-2r}$ for some $r > 0$.
(A2) Recall the Karhunen–Loéve expansion $K^{1/2}X = \sum_{j\geq 1}\xi_j e_j$. There exists a constant c such that $E[\xi_j^4] \leq c(E[\xi_j^2])^2$ for all $j \geq 1$.
(A3) $\beta(t,\cdot) \in H_K$ for all $t \in [0,1]$, and $\|\beta(t,\cdot)\|_{H_K} \in L_2$ as a function of t. Furthermore, $BK^{-1/2}$ has a bounded extension on L_2 which is Hilbert–Schmidt, where the operator B is defined by $Bf = \int \beta(\cdot,s)f(s)ds$, $f \in L_2$.

Assumption (A1) also appeared Cai and Yuan [5]. Assumption (A2) is similar to that assumed in Hall and Horowitz [12,7]. Cai and Yuan [5] assumed that $E(\int X(t)f(t)dt)^4 \leq c(E(\int X(t)f(t)dt)^2)^2$ for all $f \in L_2$. This assumption implies (A2) which can be seen by choosing $f = K^{1/2}e_j$.

(A3) is a natural extension of the case with scalar response, where $\beta(t) \in H_K$ automatically implies $K^{-1/2}\beta \in L_2$. Strictly speaking, $BK^{-1/2}$ in (A3) is only defined on the range of $K^{1/2}$, which coincides with H_K and is a dense subset of L_2. Also, note $K^{-1/2}$ is an unbounded operator. Nevertheless, it can be shown that under the condition that $\beta(t,\cdot) \in H_K$ and $\|\beta(t,\cdot)\|_{H_K} \in L_2$, $BK^{-1/2}$ has a bounded extension on L_2. More specifically, we have the following proposition whose proof is in the Appendix.

Proposition 2. *If $\beta(t,\cdot) \in H_K$ for all $t \in [0,1]$ and $\|\beta(t,\cdot)\|_{H_K} \in L_2$ where $\|\beta(t,\cdot)\|_{H_K}$ is regarded as a function of t, then $BK^{-1/2}$ has a bounded extension on L_2.*

The risk we consider is the prediction risk $E^*\|\hat{B}(X^*) - B(X^*)\|^2$ where X^* is a copy of X independent of the training data and E^* is the expectation taken over X^*. We first present the upper bound. Although this upper bound is a direct extension of the scalar response case in form, we use a quite different proof than that of Cai and Yuan [5], heavily relying on operator theory, to deal with functional responses.

Theorem 1. *Under Assumptions* (A1)–(A3) *, and that* $\lambda \asymp n^{-2r/(2r+1)}$*, we have*

$$E^*\|\hat{B}(X^*) - B(X^*)\|^2 = O_p\left(n^{-2r/(2r+1)}\right).$$

Remark 2. By examining the proof carefully, one can actually see that the convergence is uniform in β that satisfies (A3) with $\|BK^{-1/2}\|_{HS} \le 1$ (there is nothing special about the upper bound 1, which can be replaced by any $L > 0$). We can thus actually show

$$\lim_{a\to\infty}\lim_{n\to\infty}\sup_{\beta\in L_2\times H_K, \|BK^{-1/2}\|_{HS}\le 1} P(E^*\|\hat{B}(X^*) - B(X^*)\|^2 > an^{-2r/(2r+1)}) = 0.$$

This expression is put here for easy comparison with the lower bound obtained in Theorem 2.

We now establish the lower bound. This is obtained by first reducing the problem to the scalar response model and then using a slightly different construction from that used in [5] to deal with more general ϕ. The details of the proof are contained in the Appendix.

Theorem 2. *Under Assumption* (A1) *and* (A2) *on the predictor distribution, we have, for any* $a > 0$

$$\lim_{a\to 0}\lim_{n\to\infty}\inf_{\hat{\beta}}\sup_{\beta\in L_2\times H_K, \|BK^{-1/2}\|_{HS}\le 1} P(E^*\|\hat{B}(X^*) - B(X^*)\|^2 > an^{-2r/(2r+1)}) = 1$$

where the infimum is taken over all possible estimators based on the training data (X_i, Y_i), $i = 1, \ldots, n$.

3. Numerical results

3.1. Simulations

The simulation setup is similar to that used in Cai and Yuan [5]. We consider the RKHS with kernel

$$K(s,t) = \sum_{j\ge 1}\frac{2}{(j\pi)^4}\cos(j\pi s)\cos(j\pi t),$$

and thus H_K consists of functions of the form

$$f(t) = \sum_{j\ge 1} f_j \cos(j\pi t)$$

such that $\sum_j j^4 f_j^2 < \infty$. In this case, we actually have $\|f\|_{H_K}^2 = \int (f'')^2$. Data are generated from (1) without the intercept term, with

$$\beta(t,s) = \sum_j 4\sqrt{2}(-1)^j \frac{\sin(j\pi t)}{j^2}\cos(j\pi s).$$

For the covariance kernel, we use

$$\Gamma(s,t) = \sum_{j\ge 1} 2\theta_j \cos(j\pi s)\cos(j\pi t),$$

where $\theta_j = (|j - j_0| + 1)^{-2}$. When $j_0 = 1$, the two kernels are perfectly aligned, in the sense that they have the same sequence of eigenfunctions when ordered according to the eigenvalues. As j_0 increases, the level of mis-alignment also increases and we expect that the performance of functional PCA approach deteriorates with j_0. After finding the integral $Z_i(t) := \int \beta(t,s)X_i(s)ds$ (approximated easily by a Riemannian sum), we discretize Z_i over [0, 1] on an equally-spaced grid $(t_1, \ldots, t_{100})$ with 100 points and then add independent $\epsilon_{ik} \sim N(0, \sigma^2)$ noises to finally obtain $Y_i(t_k) = Z_i(t_k) + \epsilon_{ik}$. The discretized data for model fitting contains $(X_i(t_k), Y_i(t_k))$, $k = 1, \ldots, 100$, $i = 1, \ldots, n$. We set $n = 50, 100$ and $\sigma = 0.1, 0.3$, resulting in a total of four scenarios for each j_0. For values of j_0, we use $j_0 \in \{1, 3, 5, \ldots, 15\}$. For the functional PCA approach, the tuning parameter is the truncation point which we consider in the range from 1 to 25. For the RKHS approach, the tuning parameter is λ and we consider $\lambda \in \exp\{-20, -19, \ldots, 0\}$. The experiment for each scenario was repeated 100 times.

In this simulation, the tuning parameters are chosen to yield the smallest error to reflect the best achievable performance for both methods. To assess the performance, 100 test predictors $X_1^*, \ldots, X_{100}^*$ are generated from the same model as the

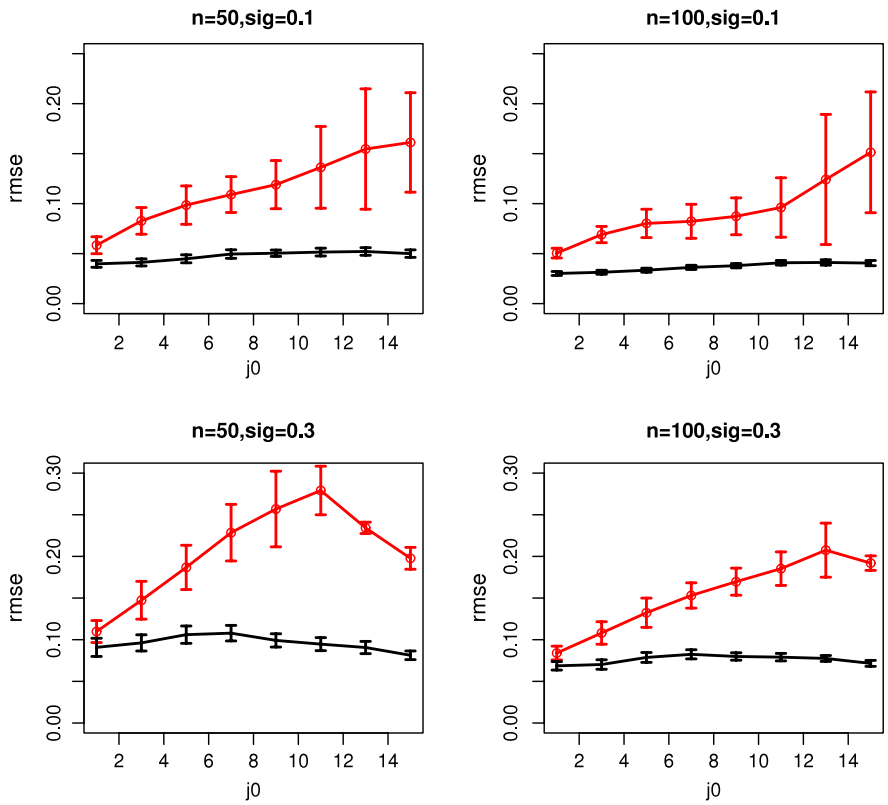

Fig. 1. RMSE for both the functional PCA method (red curve) and the RKHS method (black curve) for the simulated data using the optimal tuning parameters. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

training data, and root mean squared error (RMSE) is defined to be $(\sum_i \| \int \hat{\beta} X_i^* - \int \beta X_i^* \|^2 / 100)^{-1/2}$. Simulation results are summarized in Fig. 1, which shows the RMSE for both methods. Each panel corresponds to a pair of values of (n, σ), and the curves show the RMSE averaged over 100 replications for both methods as j_0 increases (red curve for the functional PCA approach and black curve for the RKHS approach). The vertical bar shows ± 2 standard errors computed from the 100 replications.

It is clearly seen that the performance of the RKHS approach is similar to (actually better than) that of the functional PCA approach for $j_0 = 1$. As j_0 increases, the performance of the functional PCA approach becomes much worse, while the errors for the RKHS approach remain at the same level. The difference in performance between these two methods generally increases with j_0 (with some exceptions in our particular simulations). Finally, we note that in simulation the RKHS to which the coefficient function belongs is known and the correct RKHS is used in modeling. In practice we probably will not expect such dramatic improvement as in the simulations reported above.

3.2. Real data

We now turn to the prediction performance of the proposed method on two real datasets. These datasets are used frequently in functional data analysis, and both are available from the fda package in R.

Canadian weather data. The daily weather data consists of daily temperature and precipitation measurements recorded in 35 Canadian weather stations. Each observation consists of functional data observed on an equally-spaced grid of 365 points. We treat the temperature as the independent variable and the goal is to predict the corresponding precipitation curve given the temperature measurements. As is previously done, we set the dependent variable to be the log-transformed precipitation measurements, and a small positive number is added to the values with 0 precipitation recorded. Given the periodic nature of the data, we set $H_K = \mathcal{W}_2^{per}$, the second-order Sobolev space of periodic functions on [0, 1] (see for example Chapter 2.1 of Wahba [17] for details). The reproducing kernel is given by $K(s,t) = K_1(s,t) + K_2(s,t)$ with $K_1(s,t) = 1, K_2(s,t) = \sum_{j\geq 1} \frac{2}{(2\pi j)^4} \cos(2\pi j(s-t))$. The modification as mentioned in Remark 1 is used. We use leave-one-out cross-validation to determine the best tuning parameters to use for both methods. The left panel in Fig. 2 shows the prediction errors on the 35 stations using the best tuning parameters (trained on 34 stations). For 20 stations, the functional PCA approach has larger error than the RKHS approach. The average mean prediction error for the functional PCA approach is 0.43 while the error is 0.40 for the RKHS approach.

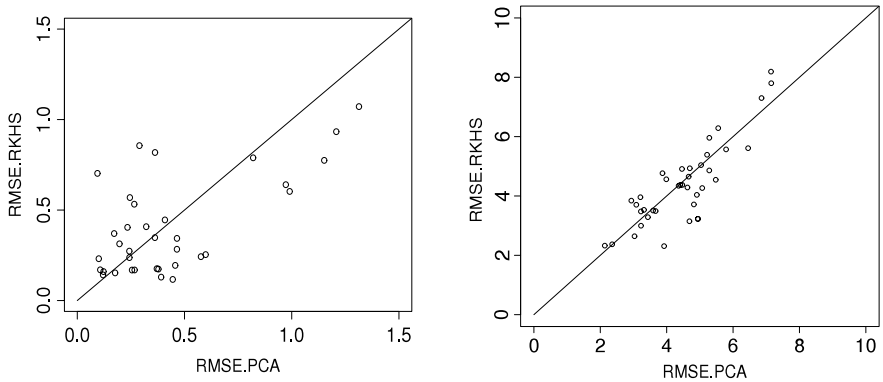

Fig. 2. Leave-one-out prediction error for the real data. The x-coordinate for each point shows the error of the functional PCA method, and the y-coordinate shows the error of the RKHS method. Left panel: Canadian weather data; Right panel: Gait data. The tuning parameters are chosen to minimize the leave-one-out cross-validation error in both methods.

Gait data. The Motion Analysis Laboratory at Children's Hospital, San Diego, collected these data, which consist of the angles formed by the hip and knee of 39 children over each child's gait cycle. The cycle begins and ends at the point where the heel of the limb under observation strikes the ground. Both sets of functions are periodic and it is of interest to see how the two joints interact. In this application, we use hip angle as the predictor and knee angle as the response. The right panel in Fig. 2 shows the prediction errors on the 39 children. For 21 children, the functional PCA approach has larger error than the RKHS approach. The average mean prediction error for the functional PCA approach is 4.49 while the error is 4.38 for the RKHS approach.

4. Conclusion

In this paper, we established the minimax rate of convergence for prediction in functional response models in the setting where the covariance kernel Γ and the reproducing kernel K are not aligned. Our simulations show that as the degree of alignment of the two kernels decreases, the RKHS estimator can significantly outperform the estimator based on functional PCA. The two real datasets further demonstrate that the RKHS estimator can have better prediction accuracy.

Choice of tuning parameter λ can be done via cross-validation, as illustrated in our analysis of the real data. Cai and Yuan [5] proposed an adaptive method for tuning parameter selection which is an important theoretical development, but in our experience does not work as well as cross-validation. Theoretical development of a good tuning parameter selector can be of significant importance which we do not investigate here.

Furthermore, one naturally wonders whether a similar RKHS approach can be extended to sufficient dimension reduction such as functional sliced inverse regression (SIR), which was also traditionally based on functional PCA which assumes that the projection direction of interest is well-represented by the basis obtained from functional PCA. It is interesting to see whether the more general framework can lead to better performance in functional SIR.

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Appendix. Proofs

Proof of Proposition 2. Let $\{\omega_j\}_{j=1}^{\infty}$ be the eigenfunctions of K corresponding to the eigenvalues $\alpha_1 \geq \alpha_2 \geq \cdots > 0$. Since $\beta(t,\cdot) \in H_K$, we can write $\beta(t,s) = \sum_j a_j(t)\omega_j(s)$ for some function a_j, with $\sum_{j=1}^{\infty} a_j^2(t)/\alpha_j < \infty$ (pointwise summable in t). For any $f = \sum_j f_j\omega_j \in H_K$, $BK^{-1/2}f = \sum_j (f_j/\sqrt{\alpha_j})a_j$. Using this representation, $BK^{-1/2}$ can be naturally extended to L_2 by defining $BK^{-1/2}f = \sum_j (f_j/\sqrt{\alpha_j})a_j \in L_2$ for any $f \in L_2$. Using Cauchy–Schwarz inequality, this operator is obviously bounded on L_2 since the assumption that $\|\beta(t,\cdot)\|_{H_K} \in L_2$ implies $(\sum_j a_j^2/\alpha_j)^{1/2} \in L_2$. $\square$

Proof of Theorem 1. In the proofs we use C to denote a generic positive constant. Using $\beta(t,\cdot) = K^{1/2}f(t,\cdot)$, from (4),

$$\hat{B}(X^*) = \frac{\sum_i (Y_i \otimes K^{1/2}X_i)}{n}(T_n + \lambda I)^{-1}K^{1/2}X^*,$$

where I is the identity operator, $T_n = K^{1/2}\Gamma_n K^{1/2}$ and $\Gamma_n = \sum_i (X_i \otimes X_i)/n$ is the empirical version of Γ. Using $Y_i = B(X_i) + \epsilon_i$, and noting that $T_n = \sum_i (K^{1/2}X_i \otimes K^{1/2}X_i)/n$, we have

$$\begin{aligned}
\hat{B}(X^*) - B(X^*) &= \frac{\sum_i (B(X_i) \otimes K^{1/2}X_i)}{n}(T_n + \lambda I)^{-1}K^{1/2}X^* + \frac{\sum_i (\epsilon_i \otimes K^{1/2}X_i)}{n}(T_n + \lambda I)^{-1}K^{1/2}X^* - B(X^*) \\
&= BK^{-1/2}\left(T_n(T_n + \lambda I)^{-1} - I\right)K^{1/2}X^* + \frac{\sum_i (\epsilon_i \otimes K^{1/2}X_i)}{n}(T_n + \lambda I)^{-1}K^{1/2}X^* \\
&= -\lambda BK^{-1/2}(T_n + \lambda I)^{-1}K^{1/2}X^* + \frac{\sum_i (\epsilon_i \otimes K^{1/2}X_i)}{n}(T_n + \lambda I)^{-1}K^{1/2}X^* \\
&=: A_1 + A_2.
\end{aligned}$$

We first deal with A_1. Note $A_1 = -\lambda BK^{-1/2}(T + \lambda I)^{-1}K^{1/2}X^* - \lambda BK^{-1/2}(T_n + \lambda I)^{-1}(T - T_n)(T + \lambda I)^{-1}K^{1/2}X^*$.

Using the expansion $K^{1/2}X^* = \sum_j \xi_j^* e_j$, we have $\lambda^2 E^*\|BK^{-1/2}(T+\lambda I)^{-1}K^{1/2}X^*\|^2 = \lambda^2 E^*\left[\sum_k \langle BK^{-1/2}\sum_j \frac{\xi_j^*}{s_j+\lambda}e_j, e_k\rangle^2\right] = \lambda^2 \sum_{j,k} \frac{s_j}{(s_j+\lambda)^2}\langle BK^{-1/2}e_j, e_k\rangle \le \frac{\lambda}{4}\sum_{j,k}\langle BK^{-1/2}e_j, e_k\rangle^2 = \frac{\lambda}{4}\|BK^{-1/2}\|_{HS}^2$.

Also, writing $\mathcal{A} = BK^{-1/2}(T_n + \lambda I)^{-1}(T - T_n)(T + \lambda I)^{-1}$ for simplicity of notation,

$$\begin{aligned}
&\lambda^2 E^*\|BK^{-1/2}(T_n + \lambda I)^{-1}(T - T_n)(T + \lambda I)^{-1}K^{1/2}X^*\|^2 \\
&= \lambda^2 E^*\langle \mathcal{A}K^{1/2}X^*, \mathcal{A}K^{1/2}X^*\rangle \\
&= \lambda^2 E^*\langle \mathcal{A}^{\mathrm{T}}\mathcal{A}K^{1/2}X^*, K^{1/2}X^*\rangle \\
&= \lambda^2 \mathrm{Trace}(\mathcal{A}^{\mathrm{T}}\mathcal{A}T) \\
&= \lambda^2 \|\mathcal{A}T^{1/2}\|_{HS}^2 \\
&\le \lambda^2 \|BK^{-1/2}\|_{HS}^2 \|(T_n + \lambda I)^{-1}(T - T_n)(T + \lambda I)^{-1}T^{1/2}\|_{HS}^2 \\
&\le \lambda^2 \|BK^{-1/2}\|_{HS}^2 \|(T_n + \lambda I)^{-1}\|_{op}^2 \|(T - T_n)(T + \lambda I)^{-1}T^{1/2}\|_{HS}^2 \\
&= O_p(\|(T - T_n)(T + \lambda I)^{-1}T^{1/2}\|_{HS}^2).
\end{aligned} \tag{5}$$

We have $E\|(T - T_n)(T + \lambda I)^{-1}T^{1/2}\|_{HS}^2 = E\sum_{j,k}\langle (T - T_n)(T + \lambda I)^{-1}T^{1/2}e_j, e_k\rangle^2 = E\sum_{j,k}\langle (T - T_n)\frac{s_j^{1/2}}{(s_j+\lambda)}e_j, e_k\rangle^2$. Direct calculation reveals that $E\langle (T - T_n)e_j, e_k\rangle^2 = E\langle s_j e_j - \frac{1}{n}\sum_i((\sum_l \xi_{il}e_l) \otimes (\sum_m \xi_{im}e_m))e_j, e_k\rangle^2 = E\langle s_j e_j - \sum_i \frac{\sum_l \xi_{il}\xi_{ij}e_l}{n}, e_k\rangle^2 = E(s_j I\{j = k\} - \frac{\sum_i \xi_{ij}\xi_{ik}}{n})^2 \le E(\frac{\sum_i \xi_{ij}\xi_{ik}}{n})^2$, where the last step used the fact that $E[\xi_{ij}\xi_{ik}] = s_j I\{j = k\}$. Using Assumption (A2), we have $E\langle (T - T_n)e_j, e_k\rangle^2 \le Cs_j s_k/n$, which then implies

$$E\|(T - T_n)(T + \lambda I)^{-1}T^{1/2}\|_{HS}^2 \le \frac{C}{n}\sum_{j,k}\frac{s_j^2 s_k}{(s_j + \lambda)^2} = O\left(\frac{1}{n}\sum_j \frac{s_j^2}{(s_j + \lambda)^2}\right). \tag{6}$$

Combining the results above we get $E^*\|A_1\|^2 = O_p(\lambda + \frac{1}{n}\sum_j \frac{s_j^2}{(s_j+\lambda)^2})$.

Now, write $A_2 = \frac{\sum_i (\epsilon_i \otimes K^{1/2}X_i)}{n}(T + \lambda I)^{-1}K^{1/2}X^* - \frac{\sum_i (\epsilon_i \otimes K^{1/2}X_i)}{n}(T + \lambda I)^{-1}(T - T_n)(T_n + \lambda I)^{-1}K^{1/2}X^*$. We have

$$E^*\left\|\frac{\sum_i (\epsilon_i \otimes K^{1/2}X_i)}{n}(T + \lambda I)^{-1}K^{1/2}X^*\right\|^2 = E^*\left\|\frac{1}{n}\sum_i \epsilon_i\left\langle K^{1/2}X_i, \sum_j \frac{\xi_j^*}{s_j + \lambda}e_j\right\rangle\right\|^2,$$

and thus $E\|\frac{\sum_i (\epsilon_i \otimes K^{1/2}X_i)}{n}(T + \lambda I)^{-1}K^{1/2}X^*\|^2 = \frac{\sigma_\epsilon^2}{n}E\left[\left\langle K^{1/2}X_1, \sum_j \frac{\xi_j^*}{s_j+\lambda}e_j\right\rangle^2\right] = \frac{\sigma_\epsilon^2}{n}E\sum_j \frac{\xi_{1j}^2\xi_j^{*2}}{(s_j+\lambda)^2} = \frac{\sigma_\epsilon^2}{n}\sum_j \frac{s_j^2}{(s_j+\lambda)^2}$, where $\sigma_\epsilon^2 = E\|\epsilon\|^2$.

Furthermore, denoting $\mathcal{C} = (T + \lambda I)^{-1}(T - T_n)(T_n + \lambda I)^{-1}$,

$$\begin{aligned}
&E\left[\left\|\frac{\sum_i (\epsilon_i \otimes K^{1/2}X_i)}{n}(T_n + \lambda I)^{-1}(T - T_n)(T + \lambda I)^{-1}K^{1/2}X^*\right\|^2 |X_1, \ldots, X_n\right] \\
&= \frac{\sigma_\epsilon^2}{n^2}E\left[\sum_i \langle K^{1/2}X_i, (T_n + \lambda I)^{-1}(T - T_n)(T + \lambda I)^{-1}K^{1/2}X^*\rangle^2 |X_1, \ldots, X_n\right]
\end{aligned}$$

$$
\begin{aligned}
&= \frac{\sigma_\epsilon^2}{n^2} E\left[\sum_i \langle \mathcal{C}K^{1/2}X_i, K^{1/2}X^*\rangle^2 | X_1, \ldots, X_n\right] \\
&= \frac{\sigma_\epsilon^2}{n^2}\left[\sum_i \langle \mathcal{C}^{\mathrm{T}}T\mathcal{C}K^{1/2}X_i, K^{1/2}X_i\rangle\right] \\
&= \frac{\sigma_\epsilon^2}{n}\mathrm{Trace}(\mathcal{C}^{\mathrm{T}}T\mathcal{C}T_n) \\
&= \frac{\sigma_\epsilon^2}{n}\mathrm{Trace}(T_n^{1/2}\mathcal{C}^{\mathrm{T}}T^{1/2}T^{1/2}\mathcal{C}T_n^{1/2}) \\
&= \frac{\sigma_\epsilon^2}{n}\|T_n^{1/2}\mathcal{C}^{\mathrm{T}}T^{1/2}\|_{HS}^2 \\
&= \frac{\sigma_\epsilon^2}{n}\|T_n^{1/2}(T_n+\lambda I)^{-1}(T-T_n)(T+\lambda I)^{-1}T^{1/2}\|_{HS}^2 \\
&\le \frac{\sigma_\epsilon^2}{n}\|T_n^{1/2}(T_n+\lambda I)^{-1}\|_{op}^2\|(T-T_n)(T+\lambda I)^{-1}T^{1/2}\|_{HS}^2 \\
&= O_p\left(\frac{1}{n\lambda}\right)\cdot O_p\left(\frac{1}{n}\sum_j \frac{s_j^2}{(s_j+\lambda)^2}\right) \\
&= o_p\left(\frac{1}{n}\sum_j \frac{s_j^2}{(s_j+\lambda)^2}\right),
\end{aligned}
$$

where we used (6) and that $n\lambda \to \infty$. Thus we have $E^*\|A_2\|^2 = O_p(\frac{1}{n}\sum_j \frac{s_j^2}{(s_j+\lambda)^2})$. The theorem is proved by combining the bounds for $E^*\|A_1\|^2$ and $E^*\|A_2\|^2$ and plugging in $\lambda \asymp n^{-2r/(2r+1)}$, $s_j \asymp j^{-2r}$. □

Proof of Theorem 2. Our model is $Y(t) = \int \beta(t,s)X(s)ds + \epsilon(t)$. Consider the special case $\beta(t,s) = e_1(t) \otimes \beta(s)$ and $\epsilon(t) = e_1(t)\chi$, where $\beta(s) \in H_K$, $\|\beta\|_{H_K} \le 1$, and $\chi \sim N(0,\sigma^2)$. Then by taking inner products with e_j on both sides of $Y(t) = \int \beta(t,s)X(s)ds + \epsilon(t)$, the model becomes $Y^{(1)} = \int \beta(s)X(s)ds + \chi$, $Y^{(2)} = Y^{(3)} = \cdots = 0$ where $Y^{(j)} = \langle Y, e_j\rangle$. Since $\|\int \beta(\cdot,s)X(s)ds\| = |\int \beta(s)X(s)ds|$, the lower bound for the scalar response model provides a lower bound for the functional response model. Thus our lower bound follows from the result in [5]. □

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